Waves in dispersive media

F.T. $\begin{cases} f(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t} \tilde{f}(\omega) \\ \tilde{f}(\omega) = \int dt\, e^{i\omega t} f(t) \end{cases}$

$$\Rightarrow \frac{\partial}{\partial t} f(t) = \int \frac{d\omega}{2\pi} (-i\omega) e^{-i\omega t} \tilde{f}(\omega)$$

F.T. of Maxwell's eqns in media gives:

$$\begin{cases} \vec{\nabla} \cdot \tilde{D} = \tilde{\rho}_f \\ \vec{\nabla} \cdot \tilde{B} = 0 \\ \vec{\nabla} \times \tilde{E} - i\omega \tilde{B} = 0 \\ \vec{\nabla} \times \tilde{H} + i\omega \tilde{D} = \tilde{J}_f \end{cases}$$

so that only spatial derivatives. All entries are F.T. of t-dependent fields

$\tilde{D} \equiv \tilde{D}(\omega)$, $\tilde{H} \equiv \tilde{H}(\omega)$ etc.

Linear Medium $\begin{cases} \tilde{P}(\omega) = \varepsilon_0 \chi_e(\omega) \tilde{E}(\omega) \\ \tilde{M}(\omega) = \chi_m(\omega) \tilde{H}(\omega) \end{cases}$

$\tilde{D} = \varepsilon_0 \tilde{E} + \tilde{P} = \varepsilon_0 \varepsilon_r(\omega) \tilde{E} = \varepsilon(\omega) \tilde{E}$

$\tilde{H} = \frac{1}{\mu_0} \tilde{B} - \tilde{M} \Rightarrow \tilde{B} = \mu_0 \mu_r(\omega) \tilde{H}(\omega)$

$\underline{\tilde{B} = \mu(\omega) \tilde{H}}$

Assume <u>uniform media</u> $\Rightarrow \vec{\nabla} \cdot (\varepsilon \tilde{E}) = \varepsilon \vec{\nabla} \cdot \tilde{E}$

(i.e. spatial derivatives do not act on $\varepsilon(\omega), \mu(\omega)$)

Then $\begin{cases} \vec{\nabla} \cdot \tilde{E} = \frac{\tilde{\rho}_f}{\varepsilon} \\ \mu \vec{\nabla} \cdot \tilde{H} = 0 \end{cases}$ $\begin{cases} \vec{\nabla} \times \tilde{E} - i\omega \mu(\omega) \tilde{H} = 0 \\ \vec{\nabla} \times \tilde{H} + i\omega \varepsilon(\omega) \tilde{E} = \tilde{J}_f \end{cases}$

Transverse wave

No external sources/ currents

$$\vec{\nabla} \times \vec{H} + i\omega \varepsilon(\omega) \vec{E} = 0$$

$$\underbrace{\vec{\nabla} \times (\vec{\nabla} \times \vec{H})}_{\vec{\nabla}(\underset{\overset{\parallel}{0}}{\vec{\nabla} \cdot \vec{H}}) - \vec{\nabla}^2 \vec{H}} = -i\omega \varepsilon(\omega) \vec{\nabla} \times \vec{E}$$

but $\vec{\nabla} \times \vec{E} = i\omega \mu(\omega) \vec{H}$

$$\Rightarrow \quad -\nabla^2 \vec{H} = -i\omega \varepsilon(\omega) \left(i\omega \mu(\omega) \vec{H} \right)$$

$$\nabla^2 \vec{H} + \omega^2 \mu(\omega) \varepsilon(\omega) \vec{H} = 0 \qquad \text{wave equ.}$$

$$\mu(\omega) \varepsilon(\omega) = \mu_0 \varepsilon_0 \left(1 + \chi_m(\omega)\right)\left(1 + \chi_e(\omega)\right) \equiv \frac{1}{c^2} n^2(\omega)$$

$$\begin{cases} \nabla^2 \vec{H} + \dfrac{\omega^2}{c^2} n^2(\omega) \vec{H} = 0 \\ \nabla^2 \vec{E} + \dfrac{\omega^2}{c^2} n^2(\omega) \vec{E} = 0 \end{cases}$$

It is often valid to assume $\mu(\omega) \approx \mu_0$

$$n^2(\omega) \approx 1 + \chi_e(\omega) = \varepsilon_r(\omega) = \frac{\varepsilon(\omega)}{\varepsilon_0}$$

But $\varepsilon(\omega)$ is complex function in general

$$n(\omega) = \sqrt{\varepsilon_r(\omega)} = \sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0}} = n'(\omega) + i\, n''(\omega).$$

Transverse wave

$$\underbrace{\vec{\nabla}\times(\vec{\nabla}\times\tilde{H})} = \vec{\nabla}\,\overset{=0}{(\vec{\nabla}\cdot\tilde{H})} - \nabla^2\tilde{H}$$

$$\vec{\nabla}\times(-i\omega\varepsilon(\omega)\tilde{E} + \tilde{J}_f) = -i\omega\varepsilon(\omega)\underbrace{\vec{\nabla}\times\tilde{E}}_{i\omega\mu(\omega)\tilde{H}} + \vec{\nabla}\times\tilde{J}_f$$

$$\omega^2\underbrace{\varepsilon(\omega)\mu(\omega)}\,\tilde{H} + \nabla^2\tilde{H} = -\vec{\nabla}\times\tilde{J}_f$$

$$\mu(\omega)\varepsilon(\omega) = \frac{\mu_0\varepsilon_0}{} [1+\chi_e(\omega)][1+\chi_m(\omega)] =$$

$$\equiv \frac{1}{c^2}\, n^2(\omega)$$

$$n^2(\omega) = \frac{\mu(\omega)\varepsilon(\omega)}{\mu_0\varepsilon_0}, \quad \mu(\omega)\varepsilon(\omega) = \frac{n^2(\omega)}{c^2}$$

$$\underbrace{\vec{\nabla}\times(\vec{\nabla}\times\tilde{E})} = \vec{\nabla}\,\overset{=0}{(\vec{\nabla}\cdot\tilde{E})} - \nabla^2\tilde{E} = \cancel{\vec{\nabla}\left(\frac{\tilde{\rho}_f}{\varepsilon}\right)} - \nabla^2\tilde{E}$$

$$i\omega\mu(\omega)\,\vec{\nabla}\times\tilde{H} = i\omega\mu(\tilde{J}_f - i\omega\varepsilon\tilde{E})$$

$$\Rightarrow \omega^2\mu(\omega)\varepsilon(\omega)\,\tilde{E} + \nabla^2\tilde{E} = \frac{1}{\varepsilon(\omega)}\cancel{\vec{\nabla}(\tilde{\rho}_f)} - i\omega\mu(\omega)\,\tilde{J}_f$$

In the absence of free charges/currents

$$\begin{cases} n^2(\omega)\dfrac{\omega^2}{c^2}\tilde{E} + \nabla^2\tilde{E} = 0 \\[2ex] n^2(\omega)\dfrac{\omega^2}{c^2}\tilde{H} + \nabla^2\tilde{H} = 0 \end{cases}$$

Transverse wave

let $\vec{E} = (E_0 e^{i(kz-\omega t)}, 0, 0)$

$$\vec{B} = \frac{\vec{\nabla}\times\vec{E}}{i\omega} = \frac{i\vec{k}\times\vec{E}}{i\omega} = \frac{k}{\omega} E_0 e^{i(kz-\omega t)}\hat{y}.$$

$$\hat{z}\times\hat{x} = \hat{y}$$

$$\vec{B} = \mu\vec{H} \Rightarrow \vec{H} = \frac{k}{\omega\mu(\omega)} E_0 e^{i(kz-\omega t)}\hat{y}.$$

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) - i\omega\mu(\omega)\vec{\nabla}\times\vec{H} = 0$$

but $\vec{\nabla}\times\vec{H} = -i\omega\vec{D}$ (for $\vec{J}_f = 0$)

$$\vec{\nabla}\underbrace{(\vec{\nabla}\cdot\vec{E})}_{0} - \nabla^2\vec{E} - i\omega\mu(\omega)\left(-i\omega\varepsilon(\omega)\vec{E}\right) = 0$$

$$\omega^2\varepsilon(\omega)\mu(\omega)\vec{E} + \nabla^2\vec{E} = 0$$

$$\varepsilon(\omega)\mu(\omega) = \varepsilon_0\mu_0\left(1+\chi_e(\omega)\right)\left(1+\chi_m(\omega)\right) \equiv \frac{n^2(\omega)}{c^2}$$

$$\frac{\omega^2 n^2(\omega)}{c^2}\vec{E} + \nabla^2\vec{E} = 0$$

$$\Rightarrow \quad \omega^2 n^2(\omega) = c^2 k^2$$

$$k = \frac{n(\omega)\omega}{c} = (n' + in'')\frac{\omega}{c}$$

n' = refractive index

n'' describes absorption.

$$\Rightarrow \vec{E} = E_0 e^{in'\omega z/c - i\omega t} e^{-n''\omega z/c}\hat{x}$$

$$\begin{cases} \vec{E} = E_0 e^{i\frac{n'\omega}{c}\left(z - \frac{ct}{n'}\right)} e^{-n''\frac{\omega z}{c}}\hat{x} \\ \vec{B} = \frac{n'+in''}{c} E_0 e^{i\frac{n'\omega}{c}\left(z-\frac{ct}{n'}\right)} e^{-n''\frac{\omega z}{c}}\hat{y}. \end{cases}$$

Pointing vector $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$

Average over period

$$\overline{\vec{S}} = \frac{1}{T} \int_0^T dt \, \frac{1}{4\mu_0} \left(\vec{E}(r) e^{-i\omega t} + \vec{E}^*(r) e^{i\omega t} \right) \times$$
$$\times \left(\vec{B}(r) e^{-i\omega t} + \vec{B}^*(r) e^{i\omega t} \right)$$

$$= \frac{1}{T} \int_0^T dt \, \frac{1}{4\mu_0} \left(\vec{E}(r) \times \vec{B}^*(r) + \vec{E}^*(r) \times \vec{B}(r) \right)$$

$$= \frac{1}{2\mu_0} \mathrm{Re} \left\{ \vec{E}(r) \times \vec{B}^*(r) \right\}$$

In case of $\vec{E}(r) \sim e^{ik'z - k''z}$

$\vec{B}(r) \sim e^{ik'z - k''z}$

$\overline{S} \sim e^{-2k''z}$ energy flux into the medium decays with distance.

$2k'' = 2n'' \frac{\omega}{c} = \alpha$ = absorption coefficient.

$$K^2 = \frac{\omega^2}{c^2} n^2(\omega) \approx \frac{\omega^2}{c^2} \frac{\varepsilon(\omega)}{\varepsilon_0} \qquad (\mu(\omega) \approx \mu_0)$$

<u>Dielectrics</u> $\quad \dfrac{\varepsilon(\omega)}{\varepsilon_0} \approx 1 + \dfrac{\omega_p^2}{\omega_0^2 - \omega^2 - i\omega\gamma} = 1 + \dfrac{\omega_p^2(\omega_0^2 - \omega^2 + i\omega\gamma)}{(\omega_0^2-\omega^2)^2 + \omega^2\gamma^2}$

$\Rightarrow$ for $K = K' + iK''$

assuming $\vec{K}' \parallel \vec{K}''$

$$\begin{cases} K'^2 - K''^2 = \dfrac{\omega^2}{c^2}\left(1 + \dfrac{\omega_p^2(\omega_0^2-\omega^2)}{(\omega_0^2-\omega^2)^2+\gamma^2\omega^2}\right) \approx \dfrac{\omega_0^2}{c^2} \text{ for } \omega \approx \omega_0. \\ 2K'K'' = \dfrac{\omega^2}{c^2} \dfrac{\omega_p^2 \gamma \omega}{(\omega_0^2-\omega^2)^2+\omega^2\gamma^2} \approx \dfrac{\omega_p^2 \omega_0}{\gamma c^2} \text{ for } \omega \approx \omega_0 \text{ (near the resonance)} \end{cases}$$

$$K' = \sqrt{\frac{\omega_0^2}{c^2} + K''^2}$$

$$\Rightarrow \quad K''\sqrt{\frac{\omega_0^2}{c^2} + K''^2} = \frac{\omega_p^2 \omega_0}{\gamma c^2}$$

$$K''^4 + K''^2 \frac{\omega_0^2}{c^2} - \left(\frac{\omega_p^2 \omega_0}{\gamma c^2}\right)^2 = 0$$

$$K''^2 = -\frac{\omega_0^2}{c^2} \pm \sqrt{\frac{\omega_0^4}{c^4} + 4\frac{\omega_p^4 \omega_0^2}{\gamma^2 c^4}} =$$

$$= \frac{\omega_0^2}{c^2}\sqrt{1 + 4\frac{\omega_p^4}{\omega_0^2\gamma^2}} - \frac{\omega_0^2}{c^2}$$

$$\approx 2\frac{\omega_p^4}{c^2\gamma^2} \quad \text{for } \omega_p^2 \ll \omega_0\gamma.$$

$$K'' \approx \sqrt{2}\frac{\omega_p^2}{c\gamma}$$

Conductors
(low-frequency)

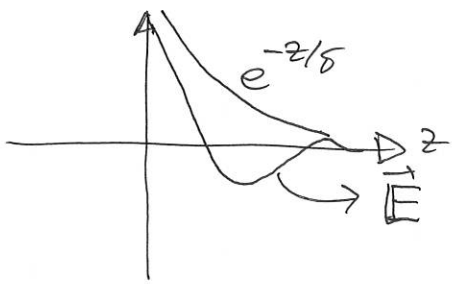
$$\frac{\varepsilon(\omega)}{\varepsilon_0} = \frac{\varepsilon_b(\omega)}{\varepsilon_0} + \frac{i\sigma_0}{\varepsilon_0\omega} \to \frac{i\sigma_0}{\omega} \quad \text{for } \omega \to 0$$

$$K^2 = \frac{\omega^2}{c^2}\frac{i\sigma_0}{\varepsilon_0\omega} \Rightarrow K = \sqrt{i\,\frac{\omega\sigma_0}{c^2\varepsilon_0}} = (1+i)\sqrt{\frac{\omega\sigma_0}{2\varepsilon_0 c^2}}$$

$$\sqrt{i} = e^{i\pi/4} = \frac{1+i}{\sqrt{2}}$$

Therefore $K' = K'' = \frac{\omega}{c}\sqrt{\frac{\sigma_0}{2\varepsilon_0\omega}} = \frac{1}{\delta}$, $\delta = \frac{c}{n''\omega}$ skin depth

$$\vec{E} = E_0\,\hat{x}\,e^{-i\omega t}\,e^{iK'z - K''z} = E_0\,\hat{x}\,e^{-\frac{z}{\delta}}\,e^{i\left(\frac{z}{\delta}-\omega t\right)}$$



$$\vec{B} = \frac{\vec{K}\times\vec{E}}{\omega} = \frac{K}{\omega}E_0\,\hat{y}\,e^{-\frac{z}{\delta}}\,e^{i\left(\frac{z}{\delta}-\omega t\right)}$$

but $\frac{K}{\omega} = \frac{K' + i\cdot K''}{\omega} = (1+i)\sqrt{\frac{\sigma_0}{2\varepsilon_0 c^2\omega}} = \frac{1+i}{\delta\omega}$

Hence observe that $\frac{c\,|B|}{|E|} = \frac{c\,|K'+iK''|}{\omega} = \frac{\sqrt{2}\,c}{\omega\delta} = \sqrt{2}\,n''$

but $n'' = \sqrt{\frac{\sigma_0}{2\varepsilon_0\omega}} \gg 1$

Magnetic field inside the conductor is much stronger than electric!

High-frequency $\omega \approx \omega_p$ $\quad \dfrac{\varepsilon(\omega)}{\varepsilon_0} = n^2(\omega) = \dfrac{\varepsilon_b(\omega)}{\varepsilon_0} - \dfrac{\omega_p^2}{\omega^2} \approx 1 - \dfrac{\omega_p^2}{\omega^2}$

$$\omega_p^2 = \frac{ne^2}{\varepsilon_0 m}.$$

Thus for $\underline{\omega < \omega_p}$ $\quad n^2(\omega) = -\left(\dfrac{\omega_p^2}{\omega^2} - 1\right) < 0$

$$\frac{c^2 k^2}{\omega^2} = 1 - \frac{\omega_p^2}{\omega^2}$$

$$c^2\left(k'^2 - k''^2 + 2ik'k''\right) = \omega^2 - \omega_p^2 < 0$$

$$\left.\begin{aligned} &\Rightarrow \quad k'^2 - k''^2 = \frac{\omega^2 - \omega_p^2}{c^2} \\ &k'k'' = 0 \end{aligned}\right\} \Rightarrow \begin{aligned} &k' = 0 \\ &k'' = \frac{\sqrt{\omega_p^2 - \omega^2}}{c} \end{aligned}$$

$$\vec{E} = E_0 \hat{x}\, e^{-i\omega t}\, e^{-k''z} \quad \text{(no propagation)}$$

$\underline{\omega > \omega_p}$: $n^2(\omega) > 0 \Rightarrow k'' = 0,\ k' = \dfrac{\sqrt{\omega^2 - \omega_p^2}}{c}$

OR, denoting $k' = k$: $\omega = \sqrt{\omega_p^2 + c^2 k^2}$

phase velocity $\longrightarrow \dfrac{\omega}{k} = \sqrt{c^2 + \dfrac{\omega_p^2}{k^2}} > c$

group velocity $v_g = \dfrac{d\omega}{dk} = \dfrac{c^2 k}{\sqrt{\omega_p^2 + c^2 k^2}} = c\,\dfrac{ck}{\sqrt{\omega_p^2 + (ck)^2}} < c$

In fact, some absorption is still present for $\omega > \omega_p$:

$$\varepsilon(\omega) = \varepsilon_b(\omega) + \frac{i\sigma(\omega)}{\omega} = \underbrace{\varepsilon_b(\omega)}_{\varepsilon_0} + \frac{i\sigma_0}{\omega(1 - i\omega\tau)}$$

$$\frac{\varepsilon(\omega)}{\varepsilon_0} = 1 + \frac{i\sigma_0}{\varepsilon_0\omega(-i)(\omega\tau+i)} = 1 - \frac{\sigma_0}{\varepsilon_0\omega^2\tau}\frac{1}{1+\frac{i}{\omega\tau}} =$$

$$= 1 - \frac{\sigma_0}{\varepsilon_0\tau\omega^2}\left(1 - \frac{i}{\omega\tau}\right) = 1 - \frac{\omega_p^2}{\omega^2} + i\frac{\omega_p^2}{\omega^3\tau}$$

Therefore $$k'^2 - k''^2 = \frac{\omega^2}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right) = \frac{\omega^2 - \omega_p^2}{c^2}$$

$$2ik'k'' = \frac{\omega^2}{c^2}\frac{\omega_p^2}{\omega^3\tau} = \frac{\omega_p^2}{c^2\omega\tau}$$

(can solve for k''.)

Group velocity

From $\omega^2 = \dfrac{c^2 k^2}{n^2(\omega)}$ can conclude that

for ω such that $n(\omega) < 1$

the propagation occurs with ~~velo~~ speed $\dfrac{c}{n(\omega)} > c$.

$$n^2(\omega) = 1 + \frac{\omega_p^2 (\omega_r^2 - \omega^2)}{(\omega_r^2 - \omega^2)^2 + \gamma^2 \omega^2} < 1 \text{ for } \omega > \omega_r \,!$$

But what matters is the <u>group velocity</u>

$$u(x,t) = \int_{-\infty}^{\infty} d\omega \; \tilde{u}(x,\omega)\, e^{i(kx - \omega t)}$$

wave packet centered at $\omega = \omega_0$

Assume
$k'' = 0$
$k' \equiv k(\omega)$

$$\Rightarrow \quad k(\omega) \cong k(\omega_0) + \frac{dk}{d\omega}(\omega - \omega_0)$$

$$u(x,t) = \cancel{\int}\; e^{i k(\omega_0) x - i \omega_0 \frac{dk}{d\omega} x} \int_{-\infty}^{\infty} d\omega \; \tilde{u}(x,\omega)\, e^{i\omega\left(\frac{dk}{d\omega} x - t\right)}$$

$$\frac{dk}{d\omega} x - t = \frac{x}{v_g} - t$$

$$v_g = \frac{d\omega}{dk} \quad \text{group velocity}$$

Longitudinal waves $\vec{k} \parallel \vec{E}$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) - i\omega\mu(\omega)\, \vec{\nabla} \times \vec{H} = 0$$

but $\vec{\nabla} \times \vec{H} = -i\omega\varepsilon(\omega)\, \vec{E}$

$$\vec{\nabla}(\vec{\nabla}\cdot\vec{E}) \overbrace{}^{-\nabla^2 \vec{E}} - \omega^2 \varepsilon(\omega)\mu(\omega)\, \vec{E} = 0$$

Try $\vec{E} = E_0 \hat{x}\, e^{ikx} \Rightarrow \vec{\nabla}\cdot\vec{E} = E_0 (ik)\, e^{ikx}$

$$\vec{\nabla}(\vec{\nabla}\cdot\vec{E}) = E_0 (ik)^2 e^{ikx}\, \hat{x}$$

and $\nabla^2 \vec{E} = \nabla^2 \left(E_0 \hat{x}\, e^{ikx}\right)$

$$= E_0 \hat{x}\, (ik)^2 e^{ikx}$$

$\Rightarrow \vec{\nabla}(\vec{\nabla}\cdot\vec{E}) - \nabla^2 \vec{E} = 0$ for such a wave.

$$\Rightarrow \omega^2 \varepsilon(\omega)\mu(\omega)\left(E_0 \hat{x}\, e^{ikx}\right) = 0$$

<u>means</u> $\varepsilon(\omega) = 0$. These are plasma waves. (or plasma oscillations)

$$\varepsilon(\omega) = \varepsilon_0 \left(1 - \frac{\omega_0^2}{\omega(\omega + \frac{i}{\tau})}\right) = 0$$

$$\omega\left(\omega + \frac{i}{\tau}\right) = \omega_0^2$$

$$\omega^2 + i\gamma\omega - \omega_0^2 = 0$$

$$\omega = \frac{-i\gamma \pm \sqrt{-\gamma^2 + 4\omega_0^2}}{2} = -\frac{i}{2\tau} + \sqrt{\omega_0^2 - \frac{1}{4\tau^2}}.$$

$$\approx \omega_0 - \frac{i}{2\tau}$$

$$\vec{E}(t) = E_0 \hat{x}\, e^{ikx}\, e^{-i\omega_0 t}\, e^{-\frac{t}{2\tau}}$$

the wave decays with time. For good conductor $\tau \sim 10^{-14}$ s.

Also, no magnetic field as $\vec{\nabla} \times \vec{E} = 0$ here.

Therefore it is simply consequence of

$$\vec{\nabla} \cdot \vec{D} = 0 \Rightarrow \varepsilon(\omega)\, \vec{\nabla} \cdot \vec{E} = 0$$

since $\varepsilon(\omega) = 0$ (but $\vec{\nabla} \cdot \vec{E} \neq 0$).

but from $\rho = \varepsilon_0\, \vec{\nabla} \cdot \vec{E}$

$$\rho = i\varepsilon_0 k E_0\, e^{ikx - i\omega_0 t}\, e^{-t/2\tau}.$$